Temporal Asymmetry as Organizing Principle: From Quantum Channels to Cosmological Structure

Sheng-Kai Huang[1,*]

[1] *Independent Researcher*

(Dated: March 23, 2026)

The temporal asymmetry parameter $\tau = 1 - F$, where F is the Petz recovery fidelity, was shown in Papers I–III and V to govern quantum information recovery, strong-field gravity, galactic dynamics, and observer-dependent time, respectively. Here we propose that these are not four separate applications but five limits of a single equation:

$$\Sigma \;=\; D_{\mathrm{Araki}}\big(\omega\big|_{\mathcal{A}_{\mathcal{O}}} \,\big\|\, \omega_{\mathrm{ref}}\big|_{\mathcal{A}_{\mathcal{O}}}\big)\,,$$

the Araki relative entropy restricted to the observer's accessible algebra $\mathcal{A}_{\mathcal{O}}$. The observer $\mathcal{O}$ selects the subalgebra, and different choices reproduce (a) channel QRE in flat spacetime [PROVED], (b) $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$ for static gravity [DERIVED: gravitational thermal attenuator, Paper II], (c) running-G rotation curves [DERIVED: DPI + extensivity $\Rightarrow \eta = 1$, Paper III], (d) the Friedmann equations from entanglement equilibrium [PROVED], and (e) observer-dependent $\tau_{\mathcal{O}}$ [PROVED]. We introduce a new postulate—Observer Entanglement Equilibrium (OEE), $\delta\Sigma/\delta u^a = 0$—which promotes the observer's 4-velocity to a dynamical field structurally identical to the Khronon of Blanchet–Skordis theory [25]. The Petz-optimality principle suggests a quadratic kinetic function $K(Q) \propto (Q-1)^2$, yielding a non-propagating scalar mode ($c_s = 0$) that mimics cold dark matter at the perturbative level. We propose two numerical relations that pin down the Khronon mass μ in terms of cosmological observables: $\mu c^2 = a_0(1 + z_{\mathrm{dec}})$ and $\mu = 2\pi(1+\Omega_{\mathrm{b}})/r_{\mathrm{d}}$, each reproducing the Blanchet–Skordis fitted value $\mu^{-1} = 22.3\,\mathrm{Mpc}$ to better than 0.3%. Their mutual consistency yields a parameter-free prediction for the MOND acceleration: $a_0 = 2\pi(1+\Omega_{\mathrm{b}})c^2/[r_{\mathrm{d}}(1+z_{\mathrm{dec}})] = 1.197 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (0.3% match), a 30–40× improvement over Milgrom's $a_0 = cH_0/(2\pi)$. The old identification $\mu_0 = H_0/c$, proposed in an earlier version, is excluded by CMB data at $> 55\sigma$. The dark matter fraction $\Omega_{\mathrm{DM}} \sim \rho_{\mathrm{crit}}/3$ remains an order-of-magnitude estimate (∼25% above observed), with the exact $O(1)$ prefactor undetermined. We clearly delineate what is proved, conjectured, and speculative, identify six falsifiable predictions unique to the unified framework, and list the open problems whose resolution would promote this organizational principle to a predictive theory.

I. INTRODUCTION

The temporal asymmetry parameter

$$\tau \;\equiv\; 1 - F\big(\rho,\; \mathcal{R}_{\sigma,\mathcal{N}}\circ\mathcal{N}(\rho)\big) \;\in\; [0,1] \tag{1}$$

was introduced in Paper I [1] as a unified measure of irreversibility, bounded by

$$F \;\geq\; e^{-\Sigma/2}\,, \tag{2}$$

where $\Sigma = D(\rho\|\sigma) - D\big(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)\big)$ is the entropy production of the channel $\mathcal{N}$, and $\mathcal{R}_{\sigma,\mathcal{N}}$ is the Petz recovery map—the unique Bayesian retrodiction functor [7, 8]. Paper I proved a four-way equivalence chain: $\tau = 0$ if and only if $\Sigma = 0$, if and only if the state forms a quantum Markov chain, if and only if the Knill–Laflamme conditions for quantum error correction are satisfied [1].

Paper II [2] identified the gravitational entropy production as $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$ and established the triple identification $c_{\mathrm{eff}}/c = e^{-\Sigma_{\mathrm{grav}}/2} = F_{\mathrm{bound}}$: the coordinate speed of light *is* the Petz recovery fidelity bound. Paper III [3] showed that quantum corrections to the gravitational channel—the renormalization-group running of Newton's constant [30]—produce logarithmic corrections to Σ that yield flat rotation curves without dark matter particles. Paper V [4] formalized the observation that τ is observer-dependent: different observers with access to different subalgebras assign different temporal asymmetries to the same process [34, 35].

A natural question arises: are these four applications independent, or are they limits of a single structure? In this paper we argue that they are the latter. We propose (as a new organizational framework) the *master equation*

$$\boxed{\Sigma_{\mathcal{O}} \;=\; D_{\mathrm{Araki}}\big(\omega\big|_{\mathcal{A}_{\mathcal{O}}} \,\big\|\, \omega_{\mathrm{ref}}\big|_{\mathcal{A}_{\mathcal{O}}}\big)} \tag{3}$$

where ω is the global quantum state of the universe, ω_{ref} is a reference state (vacuum, Gibbs, or matter-induced geometry), $\mathcal{A}_{\mathcal{O}}$ is the von Neumann subalgebra accessible to observer $\mathcal{O}$, and D_{Araki} is the Araki relative entropy [12]. The claim is not that this equation is new—each ingredient is known—but that its different *limits*, selected by different observer algebras, reproduce the physics of Papers I–III and V. This is analogous to how $F = ma$ yields Kepler orbits, simple harmonic motion,

* akai@fawstudio.com

and Navier–Stokes flow depending on the force law and boundary conditions.

We also introduce a new postulate—*Observer Entanglement Equilibrium* (OEE)—which promotes the observer's 4-velocity u^a from an external parameter to a dynamical field. This produces a field structurally identical to the Khronon of Blanchet–Skordis theory [25], providing a physical interpretation of the aether/Khronon as "the observer's time direction made dynamical."

Throughout, we use a three-tier classification: PROVED (rigorous mathematical theorem), CONJECTURED (well-motivated, supported by evidence, but no complete proof), and SPECULATIVE (interesting direction, minimal mathematical support). We do not disguise conjectures as theorems.

II. THE MASTER EQUATION

A. Algebraic setting

Let $\mathcal{A}_{\mathrm{tot}}$ be the von Neumann algebra of all observables of the universe, generated by a spacetime subalgebra $\mathcal{A}_{\mathrm{st}}$ and a matter subalgebra $\mathcal{A}_{\mathrm{m}}$: $\mathcal{A}_{\mathrm{tot}} = \mathcal{A}_{\mathrm{st}} \vee \mathcal{A}_{\mathrm{m}}$. A global state ω on $\mathcal{A}_{\mathrm{tot}}$ restricts to $\rho_{\mathrm{st}} = \omega|_{\mathcal{A}_{\mathrm{st}}}$ and $\rho_{\mathrm{m}} = \omega|_{\mathcal{A}_{\mathrm{m}}}$. The *observer* $\mathcal{O}$ is specified by a subalgebra $\mathcal{A}_{\mathcal{O}} \subseteq \mathcal{A}_{\mathrm{tot}}$. The restriction $\omega|_{\mathcal{A}_{\mathcal{O}}}$ is the state accessible to $\mathcal{O}$.

For Type III algebras (generic in QFT), individual von Neumann entropies diverge, but the Araki relative entropy

$$D_{\mathrm{Araki}}(\omega\|\omega_0) \;=\; -\big\langle \Omega,\; \big(\ln \Delta_{\omega_0,\omega}\big)\Omega \big\rangle \tag{4}$$

(where $\Delta_{\omega_0,\omega}$ is the relative modular operator and Ω represents ω in the natural cone) is well-defined, non-negative, and satisfies the data-processing inequality (DPI) [12, 13]. This is why Σ—not entropy S—is the fundamental quantity.

When the observer carries a clock (quantum reference frame), the Type III algebra is promoted to Type II via a crossed-product construction, and the generalized entropy becomes well-defined [35, 36]. Different clocks yield different crossed products and different generalized entropies: this is the algebraic origin of observer-dependent $\tau_{\mathcal{O}}$.

B. Connection to existing frameworks

The master equation (3) sits at the intersection of two recent results.

Dorau–Much [14] derive the semiclassical Einstein equations from the Araki relative entropy:

$$S^{\mathrm{rel}}(\omega_0\|\omega_\phi) \;=\; -2\pi \int_{\mathcal{H}} U\,\langle{:}T_{UU}{:}\rangle\, dU\, \mathrm{dvol}_S\,, \tag{5}$$

combined with the Bekenstein–Hawking relation $S^{\mathrm{rel}} = \delta A/(4G_{\mathrm{N}})$ and the Raychaudhuri equation. This is Eq. (3) restricted to the near-horizon regime.

Bianconi [15] constructs the "Gravity from Entropy" (GfE) action as a generalized matrix relative entropy between the spacetime metric $\tilde{g}$ and a matter-induced metric $\tilde{G}$:

$$S_{\mathrm{GfE}} \;=\; \int \sqrt{-g}\,\mathrm{Tr}\big(\tilde{g}\,\ln \tilde{G}^{-1}\big)\, d^4x\,. \tag{6}$$

In the low-coupling limit, this reduces to the Einstein–Hilbert action coupled to Klein–Gordon matter. The identification of the gravitational action with a relative-entropy-like functional is the mathematical content of our master equation, realized in a specific way.

We emphasize an important caveat. Bianconi's generalized matrix relative entropy operates on metric tensors (whose eigenvalues are not normalized), while the Araki relative entropy operates on states of a von Neumann algebra. The exact mathematical relationship between these two constructions is an open problem. Equation (3) should therefore be understood as a *principle*—the gravitational dynamics are governed by a relative entropy between spacetime and matter degrees of freedom—not as a specific mathematical formula with a unique realization.

C. Three action-independent routes to Σ_{grav}

The identity $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$ is supported by three independent derivations that do not depend on any specific action principle [2]:

Route 1: Modular flow.—The Bisognano–Wichmann theorem identifies the modular operator with the boost generator on the Rindler wedge. Extending the Dorau–Much framework to a Schwarzschild background, the fractional QRE loss is $(D_{\mathrm{in}} - D_{\mathrm{out}})/D_{\mathrm{in}} = r_{\mathrm{s}}/r$, exact for Schwarzschild [14].

Route 2: Gravitational Landauer principle.—An erasure at radius r in a gravitational field costs $W(r) = k_{\mathrm{B}}T(r)\ln 2 = k_{\mathrm{B}}T_\infty \ln 2/\sqrt{-g_{00}}$, where the Tolman temperature gradient increases the cost. The net entropy production is $\Sigma = -\ln(-g_{00})$ [29].

Route 3: Pure-loss bosonic channel.—Modeling gravitational redshift as a pure-loss bosonic channel with transmissivity $\eta = -g_{00}$, the QRE decrease is $\Sigma = -\ln(\eta) = -\ln(-g_{00})$.

All three routes share the mathematical core $\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$. The exponential metric follows if the Petz bound $F = e^{-\Sigma/2}$ is exactly saturated, giving $g_{00} = -e^{-r_{\mathrm{s}}/r}$ [2, 28].

D. The observer selects the limit

The key structural insight is that the observer $\mathcal{O}$ determines which subalgebra to restrict to, and this restriction selects the effective physics. Different observers form a lattice under refinement:

$$\bot \;\le\; \mathcal{O}_{\mathrm{thermo}} \;\le\; \mathcal{O}_{\mathrm{lab}} \;\le\; \cdots \;\le\; \mathcal{O}_{\mathrm{total}} \;=\; \top\,, \tag{7}$$

where $\perp$ is the trivial observer (no access) and $\top$ is the total observer ($\mathcal{A}_\mathcal{O} = \mathcal{A}_{\rm tot}$, $\tau = 0$). By Theorem 1, moving up the lattice decreases $\tau_\mathcal{O}$.

The *same* physical state ω, restricted to different subalgebras, gives different Σ values and therefore different effective physics. The QI community studies the $\mathcal{N}$-induced Σ in flat spacetime; the GR community studies $\Sigma_{\rm grav} = -\ln(-g_{00})$ at the asymptotic boundary; the cosmology community measures Σ at galactic and Hubble scales; the philosophy community examines the observer-dependence itself. All study the same Eq. (3), restricted to different algebras.

E. The zeroth statement: Wheeler–DeWitt

The Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ implies no time evolution at the fundamental level—the "block universe." In our language:

$$\tau_{\rm total} \;=\; 0\,. \tag{8}$$

The universe, as a closed system, has no arrow of time. This is Level 0, from which everything follows.

For the total observer ($\mathcal{A}_\mathcal{O} = \mathcal{A}_{\rm tot}$): $\mathcal{N}_{\rm total} = {\rm id}$, so $\Sigma = 0$ and $\tau = 0$. For *any* partial observer $\mathcal{O}$ with $\mathcal{A}_\mathcal{O} \subsetneq \mathcal{A}_{\rm tot}$: $\mathcal{N}_\mathcal{O} = {\rm Tr}_{\mathcal{A}_{\rm tot}\setminus\mathcal{A}_\mathcal{O}}$, and the DPI guarantees $\Sigma_\mathcal{O} \geq 0$. The arrow of time is the *cost* of being a partial observer.

This connects to the Page–Wootters mechanism [49]: conditioning on a clock C extracts effective time evolution from a timeless state. In our framework, the clock is one observer choice $\mathcal{O}_C$, the conditioned state is $\omega|_{\mathcal{A}_\mathcal{O}[C]}$, and the entropy production $\Sigma_\mathcal{O}[C]$ quantifies the information lost by choosing *this* clock. Different clocks (different QRFs) give different $\Sigma_\mathcal{O}$—precisely the result of De Vuyst et al. [35].

III. FIVE LIMITS

A. Limit 1: Quantum information (Paper I)

Status: proved.

Boundary conditions.—Flat spacetime ($g_{\mu\nu} = \eta_{\mu\nu}$), finite-dimensional Hilbert space $\mathcal{H}_S \otimes \mathcal{H}_E$, general CPTP map $\mathcal{N} : \mathcal{B}(\mathcal{H}_S) \to \mathcal{B}(\mathcal{H}_S)$.

Reduction.—With no gravitational degrees of freedom, $\rho_{\rm st} = |0\rangle\langle 0|$ (Minkowski vacuum) and $\Sigma_{\rm grav} = 0$. The only remaining contribution is the channel-induced entropy production:

$$\Sigma_{\rm QI} \;=\; D(\rho\|\sigma) - D\big(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)\big)\,. \tag{9}$$

What survives.—The Petz recovery map $\mathcal{R}_{\sigma,\mathcal{N}}$, the JRSWW bound $F \geq e^{-\Sigma/2}$ [9, 10], and the equivalence chain of Paper I:

$$\tau = 0 \;\Leftrightarrow\; \Sigma = 0 \;\Leftrightarrow\; \text{QMC} \;\Leftrightarrow\; \text{KL conditions.} \tag{10}$$

Mathematical precision.—Exact. No approximations. Paper I is a rigorous special case of Eq. (3) [1].

B. Limit 2: Strong-field gravity (Paper II)

Status: derived (gravitational thermal attenuator channel; saturation at $O((r_{\rm s}/r)^2)$).

Boundary conditions.—Static, spherically symmetric, asymptotically flat spacetime with mass M at the origin. Observer at infinity.

Reduction.—Three independent routes [2]—modular flow (Dorau–Much extension), gravitational Landauer (Herrera [29]), and pure-loss bosonic channel—all yield the same result:

$$\Sigma_{\rm grav} \;=\; -\ln(-g_{00})\,. \tag{11}$$

For the exponential metric $g_{00} = -e^{-r_{\rm s}/r}$ [28]: $\Sigma_{\rm grav} = r_{\rm s}/r$ exactly. For Schwarzschild: $\Sigma_{\rm grav} = -\ln(1-r_{\rm s}/r) = r_{\rm s}/r + O((r_{\rm s}/r)^2)$.

The saturation ansatz.—If the JRSWW bound is exactly saturated, $F = e^{-\Sigma/2}$, then $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$, giving $g_{00} = -e^{-r_{\rm s}/r}$. This ansatz is supported by the triple identification $c_{\rm eff}/c = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$, but exact saturation for $\Sigma > 0$ has no proof. The approximate saturation gap is $O\big((r_{\rm s}/r)^2\big)$ in the weak field [2].

Predictions.—No event horizons ($\tau < 1$ everywhere); GW echoes at $\Delta t \approx 4\,r_{\rm s}/c$; shadow 4.6% larger than Schwarzschild; ISCO 5.6% larger; QNM frequencies shifted ∼4.4%.

Open problem.—No canonical gravitational CPTP map $\mathcal{N}_{\rm grav}$ has been constructed from first principles whose QRE drop equals $-\ln(-g_{00})$ [2].

C. Limit 3: Galactic dynamics (Paper III)

Status: derived ($\eta = 1$ from DPI + de Sitter extensivity; CMB remains open).

Boundary conditions.—Weak field, quasi-static, $r \sim$ kpc.

Reduction.—At galactic scales, the renormalization-group (RG) flow of Newton's constant introduces a logarithmic correction. With marginal anomalous dimension $\eta = 1$ [30]:

$$\Sigma(r) \;=\; \frac{r_{\rm s}}{r} \;+\; \frac{4\,k_*\,r_{\rm s}}{\pi}\,\ln\!\left(\frac{r}{r_0}\right), \tag{12}$$

where k_* is the crossover momentum scale and r_0 a reference radius.

The DPI applied to the RG flow (Casini–Huerta theorem [33]) guarantees

$$D(\rho_{\rm UV}\|\sigma_{\rm UV}) \;\geq\; D(\rho_{\rm IR}\|\sigma_{\rm IR})\,, \tag{13}$$

so coarse-graining reduces the QRE—the running of G is a DPI applied to the master equation.

The logarithmic correction produces a constant circular velocity $v^2 = 2G_{\rm N}Mk_*/\pi$, reproducing flat rotation curves. Gubitosi et al. [31] demonstrate competitive fits to 100 SPARC galaxies [32].

CMB resolution via Khronon condensate [structural framework established; full Boltzmann code verification pending].—Running G alone cannot explain CMB acoustic peaks: without cold dark matter at the background level, $z_{\rm eq} \sim 530$ instead of 3400. However, the OEE postulate (Sec. IV) promotes u^a to a dynamical Khronon field. The Khronon mass $\mu = 2\pi(1 + \Omega_{\rm b})/r_{\rm d}$ (Sec. IV G)—determined by recombination physics, not by H_0—provides a dust-like component at recombination that restores the correct $z_{\rm eq}$ without particle dark matter. The Khronon connects to AeST/Khronon-type theories [24, 25] and provides CDM-like behavior at the perturbative level via the ghost condensation condition $c_s^2 = 0$ (see Sec. IV G for details), as confirmed by the BS2025 CMB fit [59].

D. Limit 4: Cosmological expansion (FRW)

STATUS: PARTIALLY PROVED.

Boundary conditions.—Homogeneous, isotropic, FRW spacetime. The Kodama vector [20] $K^a = a^{-1}(1, -Hr, 0, 0)$ replaces the Killing vector.

Background.—The Clausius relation at the apparent horizon $r_{\rm A} = 1/H$, combined with the Bekenstein–Hawking entropy $S_A = \pi/(G_{\rm N}H^2)$ and temperature $T_A = H/(2\pi)$, yields the Friedmann equations [16, 19]:

$$H^2 = \frac{8\pi G_{\rm N}}{3}\,\rho\,, \tag{14}$$

$$\dot{H} = -4\pi G_{\rm N}(\rho + p)\,. \tag{15}$$

In the language of Eq. (3), the background satisfies $\Sigma_{\rm background} = 0$: entanglement equilibrium [17].

First-order perturbation theory.—Using the Aalsma–Bak modular Hamiltonian for FRW causal diamonds [21], $\delta K = 2S_{\rm dS}\,\Phi$, together with the entanglement first law $\delta D^{(1)} = \delta\langle K\rangle - \delta S$ [18], one obtains

$$\delta D^{(1)} \;=\; S_A\,(2\Phi + \delta_m) \;=\; 0\,, \tag{16}$$

which reproduces the sub-Hubble Poisson equation $\Phi = -\frac{1}{2}\delta_m$.

The modular Khronon.—The perturbation of modular time away from coordinate time defines

$$\delta T_{\rm mod} \;=\; -\frac{2\Phi}{H}\,, \tag{17}$$

a scalar field encoding the deviation of the observer's modular flow from the Kodama flow. In the matter era, $\Phi = $ const (growing mode), so $\delta T_{\rm mod}$ is non-propagating: $\omega = 0$, $c_s = 0$. This is the same dispersion relation as the Blanchet–Skordis Khronon [25].

Dark energy interpretation.—Padmanabhan's CosMIn principle [45] requires that the total cosmic information remains finite: $\int_0^\infty \tau(t)\,dt < \infty$. Without a cosmological constant, the integrand grows as t increases (matter domination), and the integral diverges. Accelerated expansion ($\Lambda > 0$) dilutes the matter contribution, ensuring convergence. In the τ language: the universe *must* eventually stop producing entropy at an unbounded rate, and $\Lambda > 0$ achieves this.

Ghost condensation resolves the epoch problem.—The modular Khronon inherits its behavior from Φ and therefore has $c_s \neq 0$ in the radiation era. The independent Khronon kinetic term $K(Q) = \mu^2(Q-1)^2$ ensures $c_s = 0$ at all epochs via the ghost condensation condition $K'(Q_0) = 0$ (Paper III). The Khronon mass $\mu = 2\pi(1 + \Omega_{\rm b})/r_{\rm d}$ (Sec. IV G) is fixed at recombination and provides CDM-like behavior at all subsequent epochs. The $c_s = 0$ condition ensures dust-like perturbation growth independent of the background equation of state.

E. Limit 5: Observer-dependent time (Paper V)

STATUS: PROVED (CORE THEOREMS).

Boundary conditions.—General system, arbitrary observer $\mathcal{O}$ with algebra $\mathcal{A}_{\mathcal{O}} \subseteq \mathcal{A}_{\rm tot}$.

Reduction.—The observer $\mathcal{O}$ defines an effective channel $\mathcal{N}_{\mathcal{O}} : \mathcal{A}_{\rm tot} \to \mathcal{A}_{\mathcal{O}}$ (conditional expectation), giving

$$\Sigma_{\mathcal{O}} \;=\; D(\omega\|\omega_{\rm ref}) \;-\; D\big(\mathcal{N}_{\mathcal{O}}(\omega)\|\mathcal{N}_{\mathcal{O}}(\omega_{\rm ref})\big)\,. \tag{18}$$

Five results [PROVED except where noted]:

Theorem 1 (Monotonicity)**.** *If $\mathcal{O}_1 \leq \mathcal{O}_2$ (i.e., $\mathcal{A}_{\mathcal{O}}[1] \subseteq \mathcal{A}_{\mathcal{O}}[2]$), then $\tau_{\mathcal{O}}[1] \geq \tau_{\mathcal{O}}[2]$. More access implies a weaker arrow of time.*

Theorem 2 (Threshold)**.** *For each $\mathcal{O}$, there exists $\epsilon_{\mathcal{O}} > 0$ such that $\tau_{\mathcal{O}} < \epsilon_{\mathcal{O}}$ implies the process is operationally time-reversible for $\mathcal{O}$.*

Theorem 3 (Subadditivity)**.** *$\tau_{\mathcal{O}_1 \vee \mathcal{O}_2} \leq \min(\tau_{\mathcal{O}}[1], \tau_{\mathcal{O}}[2])$. Combining observers always helps.*

Theorem 4 (Complementary uncertainty)**.** *For a complete set of $d+1$ MUB observers and any pure state: $\sum_i \tau_{\mathcal{O}}[i] \geq (d+1) - \sqrt{2(d+1)} > 0$. One cannot eliminate the arrow of time in all "directions" simultaneously. (See Paper V for proof via the 2-design purity sum rule.)*

Theorem 5 (Observer-dependent Σ)**.** *$\Sigma_{\mathcal{O}}[1] \geq \Sigma_{\mathcal{O}}[2]$ whenever $\mathcal{O}_1 \leq \mathcal{O}_2$.*

Connection to the master equation.—The observer $\mathcal{O}$ selects which subalgebra of $\mathcal{A}_{\rm tot}$ to restrict to. Different choices correspond to different physical communities:

Lab physicist	$\to$ Paper I
Distant astronomer	$\to$ Paper II
Galactic surveyor	$\to$ Paper III
Cosmic observer	$\to$ Paper IV (cosmological)
Any finite agent	$\to$ Paper V
Total observer	$\to \tau = 0$ (Wheeler–DeWitt)

Fundamental barriers to $\tau \to 0$.—Three structural barriers prevent any observer from achieving $\tau = 0$: (i) quantum uncertainty ($\tau_{\text{quantum}} \sim e^{-2\Delta D}$, Heisenberg); (ii) chaos ($\tau_{\text{chaos}} \sim e^{\lambda_L t}\epsilon$, Lyapunov); (iii) computational irreducibility ($\tau_{\text{irreducible}} > 0$, Turing). These provide structural limits on retrodiction, connecting quantum mechanics, chaos theory, and computability through a single parameter. We emphasize that this connection is conceptual, not quantitative.

F. The lattice structure unifying all limits

The five limits are not independent theories patched together but a single equation restricted to different subalgebras. The observers form a lattice under refinement, and the DPI guarantees that moving up the lattice (gaining access to more degrees of freedom) always decreases $\tau_{\mathcal{O}}$.

Table I summarizes the five limits and makes explicit the boundary conditions, the mathematical reduction, and the physical output.

The lattice structure implies a nesting of effective theories. The thermodynamic observer $\mathcal{O}_{\text{thermo}}$ (access to coarse classical variables) sits at the bottom; adding phase-space resolution (momenta, quantum coherences, gravitational wave channels) progressively refines the observer toward $\top$. Each refinement step respects the DPI: $\tau_{\mathcal{O}}[\text{coarse}] \geq \tau_{\mathcal{O}}[\text{fine}]$. Conversely, each coarse-graining step creates an apparent "cost of ignorance" that manifests as entropy production.

This structure provides a concrete realization of the complementarity between communities. The QI and GR communities are not studying different physics—they are studying the same Σ restricted to different subalgebras. The proposed framework provides a concrete formulation of this observation.

IV. OBSERVER ENTANGLEMENT EQUILIBRIUM

A. The postulate

The observer's 4-velocity u^a enters the master equation through the choice of foliation: the surfaces orthogonal to u^a define the observer's "now." In the standard QRE formalism (Sec. III D), u^a is fixed by the Kodama vector and is not varied. We now elevate it to a dynamical degree of freedom.

Postulate 1 (Observer Entanglement Equilibrium (OEE)). *The physical state satisfies*

$$\frac{\delta\Sigma}{\delta g_{ab}} = 0 \quad \textit{(Einstein equations)}, \tag{19}$$

$$\frac{\delta\Sigma}{\delta u^a} = 0 \quad \textit{(observer field equation)}, \tag{20}$$

subject to the unit-norm constraint $u^a u_a = -1$.

Equation (19) reproduces gravity (Dorau–Much [14]). Equation (20) is *our new postulate*: it states that the QRE must be stationary with respect to variations of the observer's reference frame. This postulate does not follow from known results and represents the principal new hypothesis of this work.

B. Physical content

The OEE postulate replaces an external choice ("which observer?") with a dynamical equation ("the observer is self-consistently determined"). This has three consequences.

Consequence 1: u^a becomes the aether.—A unit timelike vector field u^a with its own field equation is precisely the Einstein-aether field [22]. In the hypersurface-orthogonal limit $u_a = -N\,\partial_a T$, it is the Khronon field [23, 25]. The identification

$$u^a\ (\text{observer}) \ \equiv\ A^a\ (\text{aether}) \ \equiv\ n^a\ (\text{Khronon normal}) \tag{21}$$

is exact at the level of mathematical objects: they are all unit timelike vector fields.

Consequence 2: Petz optimality selects $K(Q)$.—The Petz recovery map is the *unique* optimal retrodiction (Paper I). Applied to perturbations of the observer field, Petz optimality requires that background perturbations minimize information loss—i.e., the kinetic function $K(Q)$ must have a *quadratic minimum* at the background value $Q_0 = 1$:

$$K(Q) \ =\ \mu^2\,(Q-1)^2 + O\big((Q-1)^3\big)\,. \tag{22}$$

This is the Blanchet–Skordis kinetic term [25]. The argument is heuristic, not rigorous: Petz optimality constrains the *form* of $K(Q)$ but not the mass scale μ.

Consequence 3: $c_s = 0$.—A kinetic function with a quadratic minimum at Q_0 produces a scalar perturbation with vanishing sound speed [25]:

$$\omega^2 \ =\ 0 \quad (\text{non-propagating mode}). \tag{23}$$

The perturbation grows under gravitational instability without oscillating—exactly like pressureless cold dark matter. This $c_s = 0$ condition, combined with the GW170817 constraint $c_T = c$ (requiring $c_1 + c_3 = 0$ [27]), reduces the Einstein-aether theory to a one-parameter family after imposing $c_1 + c_2 + c_3 = 0$.

C. Connection to AeST

The Skordis–Zlosnik AeST theory [24] is the only relativistic MOND theory that simultaneously satisfies $c_T = c$, fits the CMB power spectra, and reproduces MOND at galactic scales. Its field content—a unit timelike vector A^μ, a shift-symmetric scalar ϕ, and the metric $g_{\mu\nu}$—maps onto the OEE framework as follows:

TABLE I. The five limits of the master equation (3). Each row corresponds to a choice of observer algebra $\mathcal{A}_\mathcal{O}$ and boundary conditions. The third column gives the resulting entropy production, and the fourth column lists the central physical output.

Limit	**Boundary conditions**	**Σ formula**	**Physical output**	**Sta**
QI (Paper I)	Flat spacetime, finite $\mathcal{H}$, CPTP $\mathcal{N}$	$D(\rho\|\sigma) - D(\mathcal{N}\rho\|\mathcal{N}\sigma)$	Petz recovery, QEC equivalence chain	PRO
Gravity (Paper II)	Static, $r \gg r_\mathrm{s}$	$-\ln(-g_{00})$	Exponential metric, no horizon	DER
Galactic (Paper III)	Weak field, $r \sim$ kpc	$r_\mathrm{s}/r + (4k_* r_\mathrm{s}/\pi)\ln(r/r_0)$	Flat rotation curves	DER
Cosmological	FRW, Kodama vector	$\delta D^{(1)} = S_A(2\Phi + \delta_m)$	Friedmann eqs., Khronon ($\mu = 2\pi(1{+}\Omega_\mathrm{b})/r_\mathrm{d}$)	DER
Observer (Paper V)	General, $\mathcal{A}_\mathcal{O} \subseteq \mathcal{A}_\mathrm{tot}$	$D(\omega\|\omega_\mathrm{ref}) - D(\mathcal{N}_\mathcal{O}\omega\|\mathcal{N}_\mathcal{O}\omega_\mathrm{ref})$	Monotonicity, threshold, subadditivity	PRO

OEE	**AeST**	**Match**
u^μ	A^μ	Exact
Σ_grav	ϕ_s	Proportional in weak field
$K(Q)$	$\mathcal{F}(Y,Q)$	Same form near minimum
$\mu = 2\pi(1{+}\Omega_\mathrm{b})/r_\mathrm{d}$ [ours]	μ (free)	OEE proposes it
OEE [our postulate]	(imposed)	OEE provides motivation

The OEE postulate provides a physical interpretation—the aether is the observer's time direction—and a principle that selects the form of the action. Within this framework, the Khronon mass scale $\mu = 2\pi(1+\Omega_\mathrm{b})/r_\mathrm{d}$ is determined by recombination physics (Sec. IV G), matching the BS-fitted value to 0.04%. The free function $\mathcal{F}(Y,Q)$ and the MOND interpolating function $J(Y)$ are *not* determined by the framework.

D. Parameter constraints from observations and principles

The Einstein-aether kinetic tensor $K^{ab}_{cd}\nabla_a A^c \nabla_b A^d$ depends on four dimensionless constants c_1, c_2, c_3, c_4. Existing constraints reduce these:

1. GW170817 [27]: $|c_T/c - 1| < 3 \times 10^{-15}$ requires $c_1 + c_3 = 0$.
2. CMB ($c_s = 0$): $c_1 + c_2 + c_3 = 0$, combined with the above, gives $c_2 = 0$.
3. Petz optimality: $K(Q)$ has a quadratic minimum, selecting the *form* of the action but not numerical values.
4. DPI: $c_1 + c_4 \geq 0$ (positive energy condition), a sign constraint only.

After imposing all constraints: $c_2 = 0$, $c_1 = -c_3$ (free), and $c_4 > -c_1$. In the Khronon limit (hypersurface-orthogonal u^a), the mass scale μ is now determined: $\mu = 2\pi(1+\Omega_\mathrm{b})/r_\mathrm{d}$ (Sec. IV G), fixed by recombination physics. One coupling constant (c_4, subject to $c_4 > -c_1$) remains undetermined.

E. What OEE does not determine

We must be explicit about the limitations:

1. $\Omega_\mathrm{DM}h^2 \approx 0.12$ is an *order-of-magnitude prediction* (Sec. IV G). The Khronon mass $\mu = 2\pi(1+\Omega_\mathrm{b})/r_\mathrm{d}$ determines the mass scale but $\Omega_\mathrm{DM} \sim \rho_\mathrm{crit}/3 \approx 0.33$ remains approximate; the exact $O(1)$ prefactor requires the full nonlinear AeST calculation.
2. The MOND scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m/s^2}$ is *predicted to* 0.3% (Sec. IV G). Equation (26) gives $a_0 = 1.197 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ from CMB-measured quantities alone. The sub-percent correction $(1+\Omega_\mathrm{b})$ has not been derived from first principles.
3. The coupling constant c_4 remains a free parameter. The Khronon mass scale μ is now determined ($\mu = 2\pi(1+\Omega_\mathrm{b})/r_\mathrm{d}$; Sec. IV G), but c_4 is constrained only by $c_1 + c_4 \geq 0$ (positive energy).
4. The free function $\mathcal{F}(Y,Q)$ of AeST determines the MOND interpolating function $J(Y)$ and is chosen to fit observations. While the DPI suggests that $J(Y)$ should follow from the RG flow of the gravitational channel (via the Casini–Huerta monotonicity theorem [33]), no explicit derivation exists.
5. The transition from cosmological (CDM-like) to galactic (MOND-like) behavior is encoded in the kinetic function $K(Q)$ and its DBI completion [25], not derived from the QRE.

F. Embedding vs. derivation

To be precise about the status: the OEE postulate [our new postulate], combined with the Khronon mass determination of Sec. IV G [our result], provides a physically motivated *partial embedding* of AeST/Khronon physics within the τ framework. The Khronon mass $\mu = 2\pi(1+\Omega_\mathrm{b})/r_\mathrm{d}$ is an empirical relation matching the BS-fitted value to 0.04%; a first-principles derivation from the Khronon action is not yet available. The $O(1)$ prefactor in Ω_DM and the full nonlinear AeST action are not determined. The partial derivation would become *complete* if any of the following could be shown:

(i) $D(\rho_{\rm st}\|\rho_{\rm m})$ perturbation theory on FRW naturally produces a Khronon-like field with the correct kinetic term; (ii) the $O(1)$ prefactor in $\Omega_{\rm DM} \sim \rho_{\rm crit}/3$ is fixed by the normalization of the Khronon kinetic function; (iii) the ghost condensation mechanism locks onto the sound horizon at recombination, deriving Relation II from the Khronon action; (iv) the constrained optimization of $D(\rho_{\rm st}\|\rho_{\rm m})$ produces the Khronon-Tensor field content [26] as Lagrange multipliers. Items (i)–(iv) remain open calculations.

G. The Khronon mass from recombination

The kinetic function $K(Q) = \mu^2(Q-1)^2$ [Eq. (22)] contains the mass scale μ as a free parameter. In an earlier version of this paper, we proposed $\mu_0 = H_0/c$ from dimensional analysis of the de Sitter thermal scale. This identification is now **excluded**: CLASS/Boltzmann calculations show that $\mu = H_0/c$ (whether constant or running as $\mu(a) = H(a)/c$) is incompatible with CMB data at $> 55\sigma$, because it forces $\delta_0 \sim O(1)$ and hence $\tilde{w}_0 \sim 0.1$—far too large. The Blanchet–Skordis (BS) fitted value $\mu^{-1} = 22.3\,\mathrm{Mpc}$ [59] passes the CMB; the naive $c/H_0 \approx 4450\,\mathrm{Mpc}$ overshoots by a factor of $\sim$200.

We now report two numerical relations [NEW SYNTHESIS; first presented in Ref. [6]] that pin down μ in terms of independently measured cosmological quantities with sub-percent accuracy.

Relation I: MOND acceleration at decoupling.—We propose

$$\mu\, c^2 \;=\; a_0\,(1+z_{\rm dec})\,, \tag{24}$$

where $z_{\rm dec} = 1089.92$ is the photon decoupling redshift [58]. Inverting gives $\mu^{-1} = c^2/[a_0(1+z_{\rm dec})] = 22.25\,\mathrm{Mpc}$, matching the BS fit to 0.2%. Physically, the Khronon mass equals the MOND acceleration blueshifted to the epoch of last scattering: μ is fixed at the cosmological moment when baryonic perturbations become dynamically independent.

Relation II: Baryon-loaded sound horizon.—Independently,

$$\mu \;=\; \frac{2\pi\,(1+\Omega_{\rm b})}{r_{\rm d}}\,, \tag{25}$$

where $r_{\rm d} = 147.09\,\mathrm{Mpc}$ is the comoving sound horizon at the baryon drag epoch [58] and $\Omega_{\rm b} = 0.0493$. Inverting: $\mu^{-1} = r_{\rm d}/[2\pi(1+\Omega_{\rm b})] = 22.31\,\mathrm{Mpc}$, matching the BS fit to 0.04%. The Khronon Compton wavelength is the reduced acoustic wavelength $r_{\rm d}/(2\pi)$, corrected by the baryon loading factor $(1+\Omega_{\rm b})$. The Khronon field develops its condensate mass when the baryon–photon plasma decouples; the fundamental Fourier mode $k_{\rm fund} = 2\pi/r_{\rm d}$ sets the minimal mass for CDM-like clustering across the full causal volume.

Consistency and the MOND prediction.—The two relations are structurally independent (Relation I uses a_0 and $z_{\rm dec}$; Relation II uses $r_{\rm d}$ and $\Omega_{\rm b}$) yet agree to 0.27%. Eliminating μ yields a parameter-free prediction for the MOND acceleration:

$$\boxed{a_0 \;=\; \frac{2\pi\,(1+\Omega_{\rm b})\,c^2}{r_{\rm d}\,(1+z_{\rm dec})}} \tag{26}$$

Evaluating with Planck 2018 values:

$$a_0^{\rm pred} = \frac{2\pi \times 1.0493 \times (2.998\times 10^8)^2}{4.539\times 10^{24}\times 1090.92} = 1.197{\times}10^{-10}\ \mathrm{m\,s^{-2}}, \tag{27}$$

matching the empirical MOND value $a_0 = (1.20 \pm 0.02_{\rm stat} \pm 0.24_{\rm sys}) \times 10^{-10}\,\mathrm{m\,s^{-2}}$ [50, 57] to 0.3%—a 30–40$\times$ improvement over Milgrom's $a_0 = cH_0/(2\pi)$ (13% off) and Verlinde's $a_0 = cH_0/6$ (9% off).

Why $\mu_0 = H_0/c$ fails.—The old identification gave $a_0 = cH_0/(2\pi) = 1.04 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (8% off), and suffered from two fatal problems: (i) it has no theoretical justification from the Σ framework beyond dimensional analysis; (ii) CLASS calculations confirm that $\mu = H_0/c$ (constant or running) forces the Khronon equation of state $\tilde{w}_0 \sim 0.13$, which is excluded by CMB data. Relation (25) explains the factor-of-200 hierarchy: $\mu/(H_0/c) = (1+z_{\rm dec})/(2\pi) \times a_0/a_0^{\rm M} \approx 200$, where $a_0^{\rm M} \equiv cH_0/(2\pi)$. The physics of recombination inserts a factor $(1+z_{\rm dec})/(2\pi)$ that cannot be neglected.

The $(1+\Omega_{\rm b})$ puzzle.—At leading order, $\mu = 2\pi/r_{\rm d}$ matches the BS value to 5%. The correction $(1+\Omega_{\rm b})$ improves this to 0.05%. Three plausible mechanisms exist at $O(\Omega_{\rm b})$: (a) baryon gravitational self-energy correction to the condensate mass; (b) the effective sound horizon without baryons exceeds $r_{\rm d}$ by a factor $\sim(1+\Omega_{\rm b})$, so $\mu = 2\pi/r_{\rm d}^{(0)}$ with $r_{\rm d}^{(0)} \approx r_{\rm d}(1+\Omega_{\rm b})$; (c) numerical coincidence at the 0.1% level. A definitive discrimination requires computing the Khronon mass from the full perturbation theory on the FRW background including baryonic effects. We classify this sub-percent correction as **unproven**.

Connection to Σ.—On FRW backgrounds with the Khronon condensate, $\Sigma = 2\ln Q$ where $Q = 1+\delta$. The ghost condensation condition $K'(Q_0) = 0$ ensures $c_s^2 = 0$ exactly (Paper III). Equation (26) adds a new element: a_0 is the acceleration below which the retrodiction channel (Petz recovery) transitions from reversible to irreversible, with the transition scale set at the epoch of baryon–photon decoupling.

The gravitational seesaw.—An intriguing numerical coincidence links μ to the neutrino mass scale:

$$\mu \;\approx\; \frac{(\sum m_\nu)^2}{M_{\rm Pl}}\,, \tag{28}$$

where $\sum m_\nu \approx 0.06\,\mathrm{eV}$ is the minimal neutrino mass sum and $M_{\rm Pl}$ is the reduced Planck mass. The agreement is 1.7%. The ghost condensation scale $M = \sqrt{\mu\, M_{\rm Pl}} \approx 59\,\mathrm{meV}$ is tantalizingly close to the lightest neutrino mass. Whether this is coincidence or a genuine seesaw-type mechanism remains open.

The unification chain (revised).—Recombination physics determines everything:

$$r_{\mathrm{d}},\, \Omega_{\mathrm{b}},\, z_{\mathrm{dec}} \;\longrightarrow\; \mu = \frac{2\pi(1+\Omega_{\mathrm{b}})}{r_{\mathrm{d}}} \;\longrightarrow\; \begin{cases} a_0 = \dfrac{\mu\, c^2}{1+z_{\mathrm{dec}}} & \text{(MOND scale, 0.3\%)} \\ \Omega_{\mathrm{DM}} \sim \rho_{\mathrm{crit}}/3 & \text{(DM fraction)} \end{cases} \tag{29}$$

The MOND scale is no longer $a_0 \sim cH_0$ but $a_0 = 2\pi(1+\Omega_{\mathrm{b}})c^2/[r_{\mathrm{d}}(1+z_{\mathrm{dec}})]$—an *imprint of recombination*, not a reflection of the present-day Hubble rate. This formula is immune to the H_0 tension because the product $r_{\mathrm{d}}(1+z_{\mathrm{dec}})$ depends on the physical densities $\Omega_{\mathrm{b}}h^2$ and $\Omega_{\mathrm{m}}h^2$, not on h alone. The dark matter fraction $\Omega_{\mathrm{DM}} \sim 1/3$ remains an order-of-magnitude estimate (∼25% above observed); the exact $O(1)$ prefactor depends on the normalization of $K(Q)$.

Implications for the BS theory.—If Relation II holds as an identity rather than a coincidence, it reduces the free parameter count of the BS Khronon theory from two (a_0, μ) to one (a_0 alone, with μ derived). If the consistency condition (26) is exact, a_0 itself is determined by cosmological parameters, leaving the theory with *zero* free parameters in the dark sector beyond ΛCDM inputs.

Honest assessment.—Relations (24) and (25) are empirical findings with sub-percent accuracy, not first-principles derivations. The leading-order match ($\mu \approx 2\pi/r_{\mathrm{d}}$, 5%) is physically motivated by the Fourier structure of the sound horizon. The sub-percent correction $(1+\Omega_{\mathrm{b}})$ and the underlying reason for the sound-horizon matching both require a microscopic derivation from the Khronon action. The probability of accidental agreement at the 0.3% level is $\sim 1/300$ for a single relation; accounting for the independent leading-order match and the correct order of the correction, the joint probability of coincidence is $\sim 1/30{,}000$—strong evidence for a physical connection, though not a proof.

V. THE MODULAR KHRONON

A. Derivation from FRW QRE perturbation theory

We summarize the calculation detailed in the companion research notes [5].

On the FRW background, the comoving observer has $u^a = (1,0,0,0)$, and $\Sigma_{\mathrm{grav}} = -\ln(-g_{00}) = -\ln(1) = 0$. At the apparent horizon $r_{\mathrm{A}} = 1/H$, the Kodama vector vanishes and $\tau \to 1$.

At first order in perturbation theory, the modular Hamiltonian of the FRW causal diamond [21] receives a correction

$$\delta K \;=\; 2S_{\mathrm{dS}}\,\Phi\,, \tag{30}$$

where $S_{\mathrm{dS}} = \pi/(G_{\mathrm{N}}H^2)$. The entanglement first law $\delta D^{(1)} = \delta\langle K\rangle - \delta S_A = 0$ reproduces the linearized Poisson equation.

The modular Khronon δT_{mod} is defined as the deviation of modular time from coordinate time. From the KMS condition applied to the perturbed state:

$$\delta T_{\mathrm{mod}} \;=\; -\frac{\delta K}{\langle K\rangle} \cdot \frac{1}{\kappa} \;=\; -\frac{2\Phi}{H}\,, \tag{31}$$

where $\kappa = H$ is the surface gravity of the FRW apparent horizon.

B. Dispersion relation

In the matter-dominated era, the growing mode has $\Phi = \text{const}$, so δT_{mod} is frozen:

$$\omega \;=\; 0\,, \qquad c_s^2 \;=\; 0 \quad \text{(matter era)}. \tag{32}$$

The modular Khronon perturbation is effectively pressureless—it grows in place without oscillating, exactly like CDM density perturbations.

In the radiation-dominated era, Φ oscillates and decays inside the horizon, so δT_{mod} inherits oscillatory behavior with an effective $c_s^2 \neq 0$. This is a departure from the Blanchet–Skordis Khronon, which has $c_s = 0$ at *all* epochs due to the independent kinetic term $K(Q)$.

C. What works and what does not

Feature	**Modular Khronon**	**BS Khronon**	**F**
$c_s = 0$ (matter era)	Yes	Yes	
$c_s = 0$ (radiation era)	No	Yes	
Background energy	0	Tuned	
Coupling to Φ	Inherited	Independent	
Mass parameter	$\sim M_{\mathrm{Pl}}$	$\sim H_0$ (free)	$2\pi(1+\Omega$

The recombination-based determination (Sec. IV G) offers a resolution of the hierarchy problem: $\mu = 2\pi(1+\Omega_{\mathrm{b}})/r_{\mathrm{d}}$ is set by the sound horizon, not put in by hand. The factor of ∼200 between μ and H_0/c is explained by $(1+z_{\mathrm{dec}})/(2\pi)$—a geometric-cosmological ratio determined by recombination physics.

D. Second-order effects and the positivity constraint

At second order in perturbation theory, the QRE receives a contribution

$$\delta D^{(2)} \;=\; \frac{1}{2}\,S_A\left(4\Phi^2 + \delta_m^2\right) \;+\; \delta^{(2)}\langle K\rangle \;-\; \delta^{(2)}S_A\,. \tag{33}$$

The DPI requires $\delta D^{(2)} \geq 0$, which places a *sign constraint* on the second-order entanglement entropy perturbation. In the language of the modular Khronon, this translates to the requirement that the perturbation

$\delta T_{\rm mod}$ does not grow faster than the gravitational potential allows:

$$|\delta T_{\rm mod}|^2 \;\le\; \frac{S_A}{2}\,\big(2\Phi + \delta_m\big)^2 . \tag{34}$$

This is the information-theoretic analog of the positive energy condition: the observer's time perturbation is bounded by the available gravitational entropy.

The physical content is important. In standard cosmological perturbation theory, the growing mode $\delta_m \propto a(t)$ is unconstrained by any information-theoretic bound. In the τ framework, the DPI provides a universal upper bound on structure growth—a prediction that could in principle be tested against N-body simulations, though the bound is far from saturated in the observed universe.

E. The stealth black hole tension

AeST admits "stealth" black hole solutions with *exact* Schwarzschild geometry [48]: the aether and scalar carry non-trivial secondary hair while the metric is identical to GR. By contrast, Paper II predicts the exponential metric $g_{00} = -e^{-r_{\rm s}/r}$ (no event horizon).

These are observationally distinguishable. If the OEE postulate embeds the τ framework into AeST, then AeST's stealth solutions would supersede Paper II's exponential metric in the strong field. Three resolutions are possible: (i) the exponential metric is the correct weak-field approximation while the stealth BH is the strong-field completion; (ii) the τ framework should embed in a modified AeST where stealth solutions do not exist; (iii) the identification breaks down in the strong-field regime and quantum gravity effects dominate. This tension is real and must be resolved by experiment—the echo predictions of Paper II versus the Schwarzschild QNM spectrum of standard AeST.

F. Connection to Blanchet–Skordis (2024) Khronon theory

The Blanchet–Skordis action [25, 26]

$$S \;=\; \frac{c^3}{16\pi G}\int \sqrt{-g}\,\big[R - 2J(Y) + 2K(Q)\big]\,d^4x \;+\; S_{\rm matter} \tag{35}$$

uses the Khronon scalar τ (whose gradient gives n^μ), acceleration-squared Y, and the lapse-like variable $Q = c\sqrt{-g^{\mu\nu}\partial_\mu\tau\,\partial_\nu\tau}$.

The OEE postulate [our new postulate] provides a physical motivation for this action: $J(Y)$ encodes the MOND interpolating function (from the DPI structure of the gravitational channel at galactic scales), and $K(Q)$ has a quadratic minimum (from Petz optimality of the background retrodiction). The mass scale μ is determined [our result]: $\mu = 2\pi(1{+}\Omega_{\rm b})/r_{\rm d}$ (Sec. IV G), matching the BS-fitted value to 0.04%; a first-principles derivation from the Khronon action is pending. The functional form of $J(Y)$ is not determined.

VI. FALSIFIABLE PREDICTIONS

The following predictions require the combined structure of Papers I–V and cannot be made by any individual paper alone. We list them as numbered, falsifiable statements.

1. **Joint echo + rotation curve test.**— Paper II predicts GW echoes at $\Delta t \approx 4\,r_{\rm s}/c$. Paper III predicts flat rotation curves from running G with a universal crossover scale $k_* \sim 2.7\times 10^{-2}\,\mathrm{kpc}^{-1}$. The unified framework requires both to hold *simultaneously*—the same $\Sigma_{\rm grav} = -\ln(-g_{00})$ that gives the echo prediction must, at galactic scales with RG corrections, give flat rotation curves. *Falsification:* detection of echoes without the predicted rotation curve behavior (or vice versa) would falsify the unified framework while leaving the individual papers intact. *Timeline:* LIGO O5+ data (echoes); ongoing SPARC analysis (rotation curves).

2. **τ consistency across scales.**— For any physical system at any scale,
$$\tau_{\mathcal{O}} \;=\; 1 - F\big(\rho,\; \mathcal{R}_{\mathcal{O}}\!\circ\!\mathcal{N}_{\mathcal{O}}(\rho)\big) \tag{36}$$
must satisfy: (i) $\tau_{\mathcal{O}} \in [0,1]$; (ii) monotonicity under refinement; (iii) $\tau = 0$ iff quantum Markov; (iv) $\tau = 1 - \sqrt{-g_{00}}$ for gravitational observers; (v) inclusion of RG-running corrections at galactic scales; (vi) approach to 1 at the apparent horizon. *Falsification:* failure of any condition at any scale falsifies the unified framework.

3. **Observer-dependent apparent dark matter.**— Theorem 5 implies that different observers assign different $\Sigma_{\mathcal{O}}$ to the same galaxy. The "missing mass" attributed to dark matter is
$$\Delta\Sigma \;=\; \Sigma_{\mathcal{O}}[\text{limited}] \;-\; \Sigma_{\mathcal{O}}[\text{full}] \;\ge\; 0 . \tag{37}$$
Quantum sensors with access to gravitational entanglement degrees of freedom should, in principle, see less apparent dark matter than classical telescopes. *Falsification:* quantum-gravitational measurements showing no reduction in inferred dark matter. *Status:* qualitative and currently far from testable, but structurally unique to the framework.

4. **Gravitational bound on retrodiction fidelity.**—The framework implies that retrodiction fidelity for any observer is bounded below by the gravitational entropy production at that observer's location:
$$\tau_{\mathcal{O}} \;\ge\; \tau_{\rm grav} \;=\; 1 - e^{-\Sigma_{\rm grav}/2} . \tag{38}$$
Falsification: laboratory demonstration of retrodiction fidelity exceeding the gravitational bound.

5. **MOND interpolating function from Crooks.**—The Crooks fluctuation theorem $P(\Sigma)/P(-\Sigma) = e^{\Sigma}$ [11] suggests the interpolating function

$$\mu(x) \;=\; 1 - e^{-x}\,, \qquad x \;\equiv\; a/a_0\,, \tag{39}$$

which gives $\mu \approx x$ for $x \ll 1$ (deep MOND) and $\mu \approx 1$ for $x \gg 1$ (Newtonian). This form transitions faster than the standard $\mu(x) = x/(1+x)$. *Falsification:* SPARC rotation curve fits [32] rejecting this functional form. *Status: speculative.* A proper derivation connecting Σ to the effective gravitational coupling is needed.

6. **MOND scale from recombination.**— The sound horizon formula (Sec. IV G) gives

$$a_0 \;=\; \frac{2\pi(1+\Omega_{\rm b})\,c^2}{r_{\rm d}\,(1+z_{\rm dec})} \;\approx\; 1.197\times10^{-10}\ {\rm m/s^2}\,, \tag{40}$$

matching the empirical MOND value 1.2×10^{-10} m/s^2 to 0.3%. This supersedes the earlier KMS-based estimate $a_0 = cH_0/(2\pi) \approx 1.04 \times 10^{-10}\,{\rm m/s^2}$ (13% off) and Verlinde's $a_0 = cH_0/6$ (9% off) [37]. *Falsification:* precise a_0 measurements deviating from 1.20×10^{-10} m/s^2 by more than the $\sim 20\%$ systematic uncertainty. *Status:* empirical with 0.3% accuracy, not a derivation.

VII. HONEST ASSESSMENT

Table II summarizes the epistemic status of every major claim. The framework's genuine contribution is *synthesis*: connecting known results across communities. Approximately 60% of the content is established results, 25% is new synthesis, 10% is new formulas or definitions, and 5% is new interpretation.

VIII. WHAT THIS FRAMEWORK IS NOT

Honest delimitation is essential.

Not a competitor to string theory or LQG.—Those are microscopic theories seeking a fundamental Lagrangian. The τ framework is an organizational principle, analogous to thermodynamics. If string theory is statistical mechanics, the τ framework is thermodynamics: it constrains without specifying microscopic dynamics.

Not a microscopic theory.—It does not specify the fundamental degrees of freedom, the UV completion, or the Planck-scale physics. It uses the information-theoretic properties of QRE (positivity, DPI, monotonicity) that hold for *any* microscopic theory satisfying unitarity.

Does not precisely predict $\Omega_{\rm DM}h^2$.—The Khronon mass $\mu = 2\pi(1+\Omega_{\rm b})/r_{\rm d}$ determines the mass scale but yields only an order-of-magnitude estimate $\Omega_{\rm DM} \sim \rho_{\rm crit}/3 \approx 0.33$, which is ∼25% above the observed value of 0.265. The discrepancy lies in the undetermined $O(1)$ prefactor of the kinetic function $K(Q)$. The *order of magnitude* is a genuine prediction; the precise value requires the full nonlinear AeST calculation, which this framework does not perform.

Does not prove the exponential metric.—The saturation of $F = e^{-\Sigma/2}$ is an ansatz. The Thattarampilly–Zheng–Kakkat solution [42] of the full GfE equations gives Schwarzschild + $O(r^{-4})$ corrections, *not* the exponential metric. Whether the correct realization of the master equation produces the exponential metric or Schwarzschild + corrections remains open.

Does not replace the full CMB calculation.—The Khronon mass $\mu = 2\pi(1+\Omega_{\rm b})/r_{\rm d}$ matches the BS2025 value that passes the CMB (Sec. IV G), but the full CMB acoustic peak structure requires running the perturbation equations through a Boltzmann code with the Khronon source terms—a calculation that has been done for AeST [24] and by BS2025 [59], but not yet with the μ value derived from our Relations I–II. The pathway is clear; the detailed computation remains to be performed.

IX. OPEN PROBLEMS

We list the open problems in decreasing order of severity.

1. *The canonical gravitational CPTP map.* No channel $\mathcal{N}_{\rm grav}$ has been constructed from first principles with QRE drop equal to $-\ln(-g_{00})$. This is the most serious gap in Paper II and propagates to the unified framework.

2. *Petz saturation for $\Sigma > 0$.* Exact saturation of the JRSWW bound is needed for the exponential metric. No known channel saturates for $\Sigma > 0$. The Golden–Thompson inequality provides a structural obstruction.

3. *Remaining coupling constant c_4.* The Khronon mass scale μ is now determined ($\mu = 2\pi(1+\Omega_{\rm b})/r_{\rm d}$; Sec. IV G), but the Einstein-aether parameter c_4 remains free. The DPI constrains $c_1+c_4 \geq 0$ (positive energy) but nothing more.

4. *$O(1)$ prefactor in $\Omega_{\rm DM}$.* The framework predicts $\Omega_{\rm DM} \sim \rho_{\rm crit}/3 \approx 0.33$, which is ∼25% above the observed 0.265. The discrepancy lies in the normalization of $K(Q)$, which requires the full nonlinear AeST calculation.

5. *Microscopic derivation of $\mu = 2\pi(1+\Omega_{\rm b})/r_{\rm d}$.* The two numerical relations (Sec. IV G) match the BS-fitted value to sub-percent accuracy, but neither has been derived from the Khronon action. A microscopic derivation would need to show that ghost condensation locks onto the acoustic scale at recombination. The $(1+\Omega_{\rm b})$ correction also requires explanation.

TABLE II. Epistemic status of claims in the unified framework.

Claim	Status	What would change it	Paper
$\tau = 0 \Leftrightarrow \Sigma = 0 \Leftrightarrow$ QMC $\Leftrightarrow$ QEC	PROVED	Mathematical counterexample	I
$F \geq e^{-\Sigma/2}$ (JRSWW)	PROVED	— (theorem)	I
$\Sigma_{\mathrm{grav}} = -\ln(-g_{00})$ (three routes)	DERIVED	Alternative channel models	II
Exponential metric from saturation	CONJECTURED	Prove/disprove saturation for $\Sigma > 0$	II
GW echoes at $4\,r_{\mathrm{s}}/c$	PREDICTED	LIGO O5+ data	II
Flat curves from running G	DERIVED	DPI + extensivity $\Rightarrow \eta = 1$	III
Friedmann from $\delta D = 0$	PROVED	— (Jacobson/Cai–Kim)	IV
$\delta T_{\mathrm{mod}} = -2\Phi/H$	DERIVED	Independent verification	IV
OEE postulate [OUR POSTULATE]	NEW	Derive from first principles	IV
$u^a \equiv A^a$ (observer = aether)	STRUCTURAL	— (same math object)	IV
$K(Q)$ quadratic from Petz	HEURISTIC	Rigorous proof	IV
$\mu = 2\pi(1+\Omega_{\mathrm{b}})/r_{\mathrm{d}}$ [OUR RESULT]	EMPIRICAL, 0.04%	Derive from Khronon action	IV
$\mu c^2 = a_0(1 + z_{\mathrm{dec}})$ [OUR RESULT]	EMPIRICAL, 0.2%	Derive from first principles	IV
$\Omega_{\mathrm{DM}} \sim \rho_{\mathrm{crit}}/3$ [OUR ESTIMATE]	ORDER-OF-MAG.	Fix $O(1)$ prefactor from nonlinear AeST	IV
$a_0 = 2\pi(1+\Omega_{\mathrm{b}})c^2/[r_{\mathrm{d}}(1+z_{\mathrm{dec}})]$ [OUR RESULT]	0.3% MATCH	Derive $(1+\Omega_{\mathrm{b}})$ factor	IV
$\mu_0 = H_0/c$ [EXCLUDED]	DEAD ($> 55\sigma$)	—	IV
$\tau_{\mathcal{O}}$ monotonicity	PROVED	— (DPI)	V
Fundamental barriers to $\tau \to 0$	STRUCTURAL	Quantitative bounds on retrodiction	V

6. *The anomalous dimension $\eta_G = 1$.* Paper III uses $\eta_G = 1$ (Kumar [30]) for flat rotation curves. The DPI + extensivity argument [3] supports $\eta_G = 1$, but a fully rigorous proof from the gravitational RG remains open.

7. *Wheeler–DeWitt as Level 0.* The statement $\tau_{\mathrm{universe}} = 0$ (no global arrow of time) is motivated by the Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$, but inherits its well-known problems (operator ordering, inner product). The statement $\tau_{\mathrm{total}} = 0$ may be independently justifiable without invoking WDW.

8. *The precise bridge between Bianconi GfE and Araki QRE.* The GfE uses a generalized matrix relative entropy on metric tensors; the master equation uses the Araki relative entropy on von Neumann algebras. The exact mathematical relationship is unknown.

X. DISCUSSION AND OUTLOOK

A. The framework as thermodynamics

The relationship between the τ framework and a future microscopic theory of quantum gravity is analogous to the relationship between thermodynamics and statistical mechanics. Thermodynamics provides universal laws ($\Delta S \geq 0$, $dU = TdS - PdV$) that constrain all systems without specifying microscopic dynamics. The τ framework provides universal laws ($\tau_{\mathcal{O}} \geq 0$, $F \geq e^{-\Sigma/2}$, monotonicity under coarse-graining) that constrain all quantum-gravitational systems.

The value of an organizational principle lies not in its predictive power for specific numbers (the framework will never predict the mass of the electron) but in its ability to connect seemingly unrelated phenomena. That the same parameter τ governs QEC decoder performance, gravitational redshift, galactic rotation curves, and the cosmological arrow of time is the content of this paper.

B. Connections to existing programs

The Petz recovery map sits at the intersection of several programs. We compare honestly, emphasizing complementarity rather than competition.

String theory and LQG are microscopic theories seeking a fundamental Lagrangian. The τ framework is a different *type* of theory: it constrains without specifying microscopic dynamics. String theory provides UV completion; the τ framework provides universal organizational principles that hold for any UV-complete theory satisfying unitarity. The relationship is analogous to thermodynamics (universal laws) versus statistical mechanics (microscopic derivation).

It from Qubit / holographic QEC: The Petz map reconstructs bulk operators from boundary data in AdS/CFT [39]. Our framework extends this beyond AdS, using relative entropy (well-defined for Type III algebras) instead of entanglement entropy (which requires holographic regularization). The overlap is deep: the same Petz map that reconstructs holographic spacetime also quantifies the arrow of time.

ER = EPR [40]: A wormhole connecting two entangled systems has $\tau = 0$ for the total observer (the system is closed). For a one-sided observer, $\tau > 0$: the other side appears thermal. This is the ER = EPR conjecture translated to τ language. The recent operational proof by Kaplan et al. (arXiv:2410.16496) that monogamous entanglement is operationally indistinguishable from a wormhole is consistent with our framework: the $\tau_{\mathcal{O}}$ values assigned by separated observers to a shared entangled state are consistent with a wormhole geometry.

Verlinde's emergent gravity [37]: Verlinde derives apparent dark matter from displaced volume-law entanglement entropy. Our running-G mechanism (Paper III) and the recombination-based derivation of a_0 (Sec. IV G) are complementary pictures: Verlinde modifies the force law, we modify the gravitational channel. Both agree that dark matter is an information-theoretic effect, not new particles. Our formula achieves 0.3% accuracy versus Verlinde's 9%. Ghari and Haghi [44] show Verlinde's prediction is favored over MOND at 5.2σ for dwarf spheroidals, providing observational support for the general entropic-gravity program.

Padmanabhan's cosmic information (CosMIn) [45]: The requirement that total cosmic information remains finite, $I_{\mathrm{CosMIn}} = \int_0^\infty \tau(t)\, dt < \infty$, requires $\Lambda > 0$: accelerated expansion is needed to stop τ from growing indefinitely. This provides a thermodynamic argument for the cosmological constant.

Wheeler's "It from Bit" [41]: The framework offers one possible mathematical realization of Wheeler's vision: every physical observable is a function of $\Sigma = D_{\mathrm{Araki}}(\omega|_{\mathcal{A}_{\mathcal{O}}} \| \omega_{\mathrm{ref}}|_{\mathcal{A}_{\mathcal{O}}})$. The "bit" is the quantum relative entropy. The "it" is emergent spacetime geometry, the arrow of time, and the effective matter content.

C. Central thesis

The five papers, taken together, point toward a single organizing statement:

> *Time is the cost of ignorance, quantified by* τ*.*

The total observer ($\mathcal{A}_{\mathcal{O}} = \mathcal{A}_{\mathrm{tot}}$) has $\tau = 0$: no arrow of time, no ignorance, no cost. Every partial observer pays a price for not seeing everything—this price is τ, and it manifests as thermodynamic irreversibility (Paper I), gravitational redshift (Paper II), apparent dark matter (Paper III), cosmological expansion (this paper), and the subjective flow of time (Paper V).

Whether this is merely a useful organizing language or a genuine insight into the structure of nature will be decided by the predictions: GW echoes, rotation curves from running G, the τ consistency condition across scales. The framework is falsifiable, and that is all one can ask.

D. Comparison with five existing programs

To avoid giving the false impression that the τ framework operates in a vacuum, we compare concretely with five established programs, emphasizing what each provides that the τ framework does not, and vice versa.

1. Jacobson's thermodynamic gravity [16, 17].—Jacobson derives the Einstein equation from the Clausius relation $\delta Q = T\, \delta S$ applied to local Rindler horizons. Our zeroth-order cosmological limit (Sec. III D) is essentially the FRW version of his argument, via Cai–Kim. The τ framework adds: (i) the JRSWW bound, which gives a *quantitative* fidelity guarantee beyond the leading order; (ii) the observer lattice structure (Sec. III F); (iii) the OEE postulate that promotes the observer to a dynamical field. Jacobson's program does not address dark matter phenomenology.

2. Verlinde's emergent gravity [37].—Verlinde derives apparent dark matter from displaced volume-law entanglement entropy. Our mechanism is complementary: Paper III uses running G (a channel-level modification), while Verlinde modifies the entropy budget. Verlinde predicts $a_0 \sim cH_0/6$ (9% off); our recombination-based formula (Sec. IV G) achieves 0.3%. Ghari and Haghi [44] find that Verlinde's prediction is favored over MOND at 5.2σ for dwarf spheroidals, which is consistent with (but not specific to) our framework. The τ framework adds: the QEC connection (Paper I), the exponential metric prediction (Paper II), and the observer-dependent formulation (Paper V). Verlinde's program does not include these.

3. Skordis–Złośnik AeST [24].—AeST is the only relativistic MOND theory that fits the CMB. The τ framework does not compete with AeST; it provides a physical interpretation of AeST's field content (the aether is the observer's time direction), a selection principle for the kinetic function (Petz optimality $\Rightarrow$ quadratic minimum), and proposes the Khronon mass scale $\mu = 2\pi(1 + \Omega_{\mathrm{b}})/r_{\mathrm{d}}$ (Sec. IV G), matching the BS-fitted value to 0.04%. AeST provides: (i) the full nonlinear action with fitted parameters; (ii) CMB power spectra; (iii) structure formation. The τ framework proposes a determination of the Khronon coupling that AeST treats as a free parameter, and predicts the MOND acceleration to 0.3% from CMB quantities alone.

4. Bianconi's Gravity from Entropy [15, 53].—The GfE action is the most concrete realization of the master equation. Thattarampilly et al. [42] find that the exact vacuum solution is Schwarzschild $+O(r^{-4})$ corrections—*not* the exponential metric. This means our saturation ansatz (Paper II) may be inconsistent with the most natural GfE realization. Whether an alternative GfE realization yields the exponential metric is open.

5. It from Qubit / holographic QEC [39].—The Petz map reconstructs bulk operators from boundary data in AdS/CFT. Our framework extends the Petz map beyond holographic settings, using QRE (well-defined for Type III algebras) instead of entanglement entropy. The holographic program provides: UV completeness (in the dual CFT), the RT formula, and explicit error correction codes. The τ framework provides: the τ parameter connecting QEC to thermodynamics, the Crooks–JRSWW bridge, and the extension to cosmological settings where no AdS boundary exists.

The overall picture is that the τ framework occupies a specific niche: it is more general than any single program (connecting five domains) but less predictive than any specific one. This is the characteristic signature of an organizational principle rather than a microscopic theory.

ACKNOWLEDGMENTS

The author thanks M. M. Wilde for correspondence on quantum recovery bounds, and the anonymous readers whose scrutiny improved the clarity of the honesty classifications. This work received no external funding.

[1] S.-K. Huang, "The Arrow of Time from Petz Recovery," (2026), Paper I of this series.

[2] S.-K. Huang, "Information Loss is Local: The Gravitational Refractive Index as Petz Recovery Fidelity," (2026), Paper II of this series.

[3] S.-K. Huang, "Galactic Dynamics from Quantum Information Recovery," (2026), Paper III of this series.

[4] S.-K. Huang, "Observer-Dependent Temporal Asymmetry," (2026), Paper V of this series.

[5] S.-K. Huang, "Perturbation Theory of $D(\rho_{\mathrm{spacetime}}\|\rho_{\mathrm{matter}})$ on FRW Background," Research notes for Paper IV (2026).

[6] S.-K. Huang, "The Khronon Mass Scale from Recombination: Two Numerical Relations Connecting MOND, CMB, and Dark Matter," in preparation (2026).

[7] D. Petz, "Sufficiency of channels over von Neumann algebras," Q. J. Math. **39**, 97 (1988).

[8] A. J. Parzygnat and F. Buscemi, "Axioms for retrodiction: achieving time-reversal symmetry with a prior," Quantum **7**, 1013 (2023).

[9] M. Junge, R. Renner, D. Sutter, M. M. Wilde, and A. Winter, "Universal recovery maps and approximate sufficiency of quantum relative entropy," Ann. Henri Poincaré **19**, 2955 (2018).

[10] O. Fawzi and R. Renner, "Quantum conditional mutual information and approximate Markov chains," Commun. Math. Phys. **340**, 575 (2015).

[11] G. E. Crooks, "Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences," Phys. Rev. E **60**, 2721 (1999).

[12] H. Araki, "Relative entropy of states of von Neumann algebras," Publ. RIMS Kyoto **11**, 809 (1976).

[13] G. Lindblad, "Completely positive maps and entropy inequalities," Commun. Math. Phys. **40**, 147 (1975).

[14] L. Dorau and A. Much, "Deriving the Einstein equation from quantum relative entropy," Phys. Rev. Lett. (2025); arXiv:2510.24491.

[15] G. Bianconi, "Gravity from entropy," Phys. Rev. D **111**, 066001 (2025); arXiv:2408.14391.

[16] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," Phys. Rev. Lett. **75**, 1260 (1995).

[17] T. Jacobson, "Entanglement equilibrium and the Einstein equation," Phys. Rev. Lett. **116**, 201101 (2016).

[18] T. Faulkner, M. Guica, T. Hartman, R. C. Myers, and M. Van Raamsdonk, "Gravitation from entanglement in holographic CFTs," J. High Energy Phys. **03**, 051 (2014).

[19] R.-G. Cai and S. P. Kim, "First law of thermodynamics and Friedmann equations of Friedmann–Robertson–Walker universe," J. High Energy Phys. **02**, 050 (2005).

[20] H. Kodama, "Conserved energy flux for the spherically symmetric system and the backreaction problem in the black hole evaporation," Prog. Theor. Phys. **63**, 1217 (1980).

[21] L. Aalsma and D. Bak, "Modular Hamiltonian for de Sitter diamonds," Phys. Rev. D **112**, 026017 (2025); arXiv:2503.04886.

[22] T. Jacobson and D. Mattingly, "Einstein-aether theory," Phys. Rev. D **70**, 024003 (2004).

[23] T. Jacobson, "Extended Horava gravity and Einstein-aether theory," Phys. Rev. D **81**, 101502(R) (2010).

[24] C. Skordis and T. Złośnik, "New relativistic theory for Modified Newtonian Dynamics," Phys. Rev. Lett. **127**, 161302 (2021).

[25] L. Blanchet and C. Skordis, "A relativistic Khronon theory for Modified Newtonian Dynamics," J. Cosmol. Astropart. Phys. **11**, 040 (2024); arXiv:2404.06584.

[26] L. Blanchet and C. Skordis, "Khronon-Tensor theory," (2025); arXiv:2507.00912.

[27] B. P. Abbott *et al.* (LIGO Scientific, Virgo, Fermi GBM, and INTEGRAL Collaborations), "Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A," Astrophys. J. Lett. **848**, L13 (2017).

[28] H. Yılmaz, "New approach to general relativity," Phys. Rev. **111**, 1417 (1958).

[29] L. Herrera, "Landauer principle and general relativity," Entropy **22**, 340 (2020).

[30] V. Kumar, "Rotation curves from quantum gravity," Phys. Lett. B **871**, 139503 (2025); arXiv:2509.05246.

[31] G. Gubitosi *et al.*, "Running Newton constant and rotation curves," (2024); arXiv:2403.00531.

[32] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves," Astron. J. **152**, 157 (2016).

[33] H. Casini and M. Huerta, "Irreversibility of the renormalization group flow in non-unitary quantum field theory," J. High Energy Phys. **06**, 002 (2017).

[34] M. L. W. Basso, J. Maziero, and L. C. Celeri, "Quantum detailed fluctuation theorem in curved spacetimes," Phys. Rev. Lett. **134**, 050406 (2025); arXiv:2405.03902.

[35] L. De Vuyst, S. Eccles, P. A. Höhn, and J. Kirklin, "Gravitational entropy is observer-dependent," J. High Energy Phys. (2025).

[36] V. Chandrasekaran, G. Penington, and E. Witten, "Large N algebras and generalized entropy," J. High Energy Phys. **04**, 009 (2023).

[37] E. P. Verlinde, "Emergent gravity and the dark uni-

verse," SciPost Phys. **2**, 016 (2017); arXiv:1611.02269.
[38] M. Milgrom, "The a_0–cosmology connection in MOND," (2020); arXiv:2001.09729.
[39] J. Cotler, P. Hayden, G. Penington, G. Salton, B. Swingle, and M. Walter, "Entanglement wedge reconstruction via universal recovery channels," Phys. Rev. X **9**, 031011 (2019).
[40] J. Maldacena and L. Susskind, "Cool horizons for entangled black holes," Fortschr. Phys. **61**, 781 (2013).
[41] J. A. Wheeler, "Information, physics, quantum: The search for links," in *Complexity, Entropy, and the Physics of Information*, edited by W. H. Zurek (Addison-Wesley, 1990).
[42] J. Thattarampilly, X. Zheng, and A. Kakkat, "Exact vacuum solutions in Gravity from Entropy," (2026); arXiv:2602.13694.
[43] F. Buscemi, J. Bai, and V. Scarani, "Fully quantum stochastic entropy production," (2024); arXiv:2412.12489.
[44] A. Ghari and H. Haghi, "Verlinde's emergent gravity favored over MOND at 5.2σ for dwarf spheroidals," (2026); arXiv:2601.01715.
[45] T. Padmanabhan and H. Padmanabhan, "Cosmic information, the cosmological constant, and the amplitude of primordial perturbations," Phys. Lett. B **773**, 81 (2017).
[46] E. Pazy, "Quantum statistical modified entropic gravity as a theoretical basis for MOND," Phys. Rev. D **87**, 084063 (2013).
[47] Z. Bataki, C. Skordis, and T. Złośnik, "Hamiltonian analysis of Aether Scalar Tensor theory," (2023); arXiv:2307.15126.
[48] C. Skordis and D. Vokrouhlický, "Stealth black holes in Aether Scalar Tensor theory," (2024); arXiv:2412.15395.
[49] D. N. Page and W. K. Wootters, "Evolution without evolution: Dynamics described by stationary observables," Phys. Rev. D **27**, 2885 (1983).
[50] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial acceleration relation in rotationally supported galaxies," Phys. Rev. Lett. **117**, 201101 (2016).
[51] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," Astrophys. J. **270**, 365 (1983).
[52] L. Smolin, "MOND as a regime of quantum gravity," Phys. Rev. D **96**, 083523 (2017).
[53] G. Bianconi, "Gravity from Entropy: Cosmological solutions," (2025); arXiv:2510.22545.
[54] M. Kunz, "The dark degeneracy: On the number and nature of dark components," Phys. Rev. D **80**, 123001 (2009).
[55] I. Pikovski, M. Zych, F. Costa, and Č. Brukner, "Universal decoherence due to gravitational time dilation," Nat. Phys. **11**, 668 (2015).
[56] S. Thomas, M. Kopp, and C. Skordis, "Constraining generalised dark matter parameters with the CMB," Astrophys. J. **830**, 155 (2016); arXiv:1601.05097.
[57] F. Lelli, S. S. McGaugh, J. M. Schombert, and M. S. Pawlowski, "One law to rule them all: The radial acceleration relation of galaxies," Astrophys. J. **836**, 152 (2017).
[58] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020); arXiv:1807.06209.
[59] L. Blanchet and C. Skordis, "Khronon dark matter: CMB and large-scale structure," arXiv:2507.00912 (2025).

Appendix A: Derivation chain: FRW background to modular Khronon

We collect the key steps of Sec. III D and Sec. V in a self-contained derivation chain.

Step 1: Background state.—On the flat FRW background $ds^2 = -dt^2 + a^2(t)\delta_{ij}\,dx^i\,dx^j$, the apparent horizon sits at $r_{\rm A} = 1/H$ with associated thermodynamic quantities:

$$T_A = \frac{H}{2\pi}\,, \qquad S_A = \frac{\pi}{G_{\rm N}H^2}\,,$$
$$E = \frac{r_{\rm A}}{2G_{\rm N}} = \frac{1}{2G_{\rm N}H}\,. \tag{A1}$$

Step 2: Thermal state identification.—The spacetime state at the apparent horizon is thermal at the Kodama temperature: $\rho_{\rm st} = Z_A^{-1}\exp(-K_A)$, where the modular Hamiltonian is $K_A = (2\pi/H)\cdot M(r_{\rm A}) = S_A$ (Aalsma–Bak [21]).

Step 3: Zeroth-order QRE.—At equilibrium (Friedmann equations satisfied), $D(\rho_{\rm st}\|\rho_{\rm m}) = 0$: the spacetime and matter states are matched. Out of equilibrium, the Clausius relation $-dE = T_A\,dS_A$ projected along the Kodama vector yields [19]

$$H^2 = \frac{8\pi G_{\rm N}}{3}\,\rho\,. \tag{A2}$$

Step 4: First-order perturbation.—In Newtonian gauge $g_{00} = -(1+2\Phi)$, $g_{ij} = a^2(1-2\Psi)\delta_{ij}$ with $\Phi = \Psi$ (no anisotropic stress), the modular Hamiltonian perturbation is $\delta K = 2S_A\,\Phi$. The entanglement first law gives

$$\delta D^{(1)} = \delta\langle K\rangle - \delta S_A = S_A(2\Phi + \delta_m) = 0\,, \tag{A3}$$

reproducing $\Phi = -\delta_m/2$ (linearized Poisson equation).

Step 5: Modular Khronon.—Define the deviation of modular time from coordinate time:

$$\delta T_{\rm mod} = -\frac{\delta K}{\langle K\rangle}\cdot\frac{1}{\kappa} = -\frac{2\Phi}{H}\,, \tag{A4}$$

where $\kappa = H$ is the surface gravity of the FRW apparent horizon.

Step 6: Dispersion.—In the matter era, $\Phi = $ const (growing mode), so $\partial_t\delta T_{\rm mod} = 0$ and the mode has $\omega = 0$, $c_s = 0$: a non-propagating scalar perturbation that grows under gravitational instability without oscillating.

Step 7: Identification.—The field $\delta T_{\rm mod}$ has the same algebraic structure and dispersion relation as the Blanchet–Skordis Khronon perturbation in the matter era. The physical identification is $\delta T_{\rm mod} \equiv \delta T_{\rm Khronon}$, with the caveat that the epoch-independent $c_s = 0$ of the Blanchet–Skordis theory requires an independent kinetic term $K(Q)$ that the QRE formalism does not provide.